

Equations and Multiple-Angle Identities

Solve each equation for $0 \leq \theta < 2\pi$.

1) $-\sin 2\theta = \sqrt{2}\sin \theta - 2\sin 2\theta$

2) $-2\cos \theta = \sin 2\theta - \cos \theta$

3) $0 = \cos 2\theta + 3\sin \theta + 4\sin^2 \theta$

4) $\cos \theta + \cos 2\theta = 0$

5) $\cos \frac{\theta}{2} = \cos \theta + 1$

6) $\cos \theta = 2 + 3\sin \frac{\theta}{2}$

7) $3 + 2\cos \theta = 4\cos \frac{\theta}{2}$

8) $\sqrt{3}\sin \frac{\theta}{2} + \cos \theta = 1$